

CompSci 234/NetSys 210  
Advanced Topics in Networking  
**Winter 2019**

# Unit 14: Internet-of-Things Analytics

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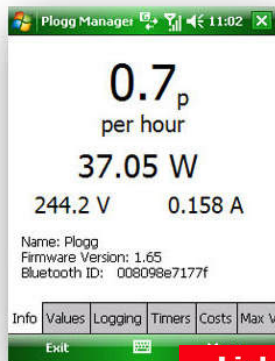
Some slide adopted from Profs. Payam Barnaghi and Chuan Fohs' materials

# Agenda

- Internet-of-Things (IoT)
- IoT Versus Cyber-Physical Systems (CPS)
- IoT Data Processing
- IoT Analytics

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# Programmable Embedded Systems Have Gradually Become Commodity



Linker Intel Group



Image Sensor Device



## Happy Birthday Raspberry Pi!

It was at the [very end of February 2012](#) that the Raspberry Pi officially went on sale for the first time. So we're coming up to the time to say "Happy Third Birthday".

# Thanks to the Networks, the “Things” are More Connected

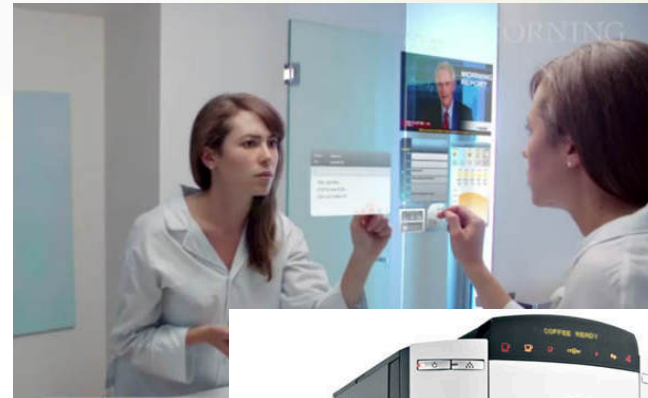
Home/daily-life devices

Business

Public infrastructure

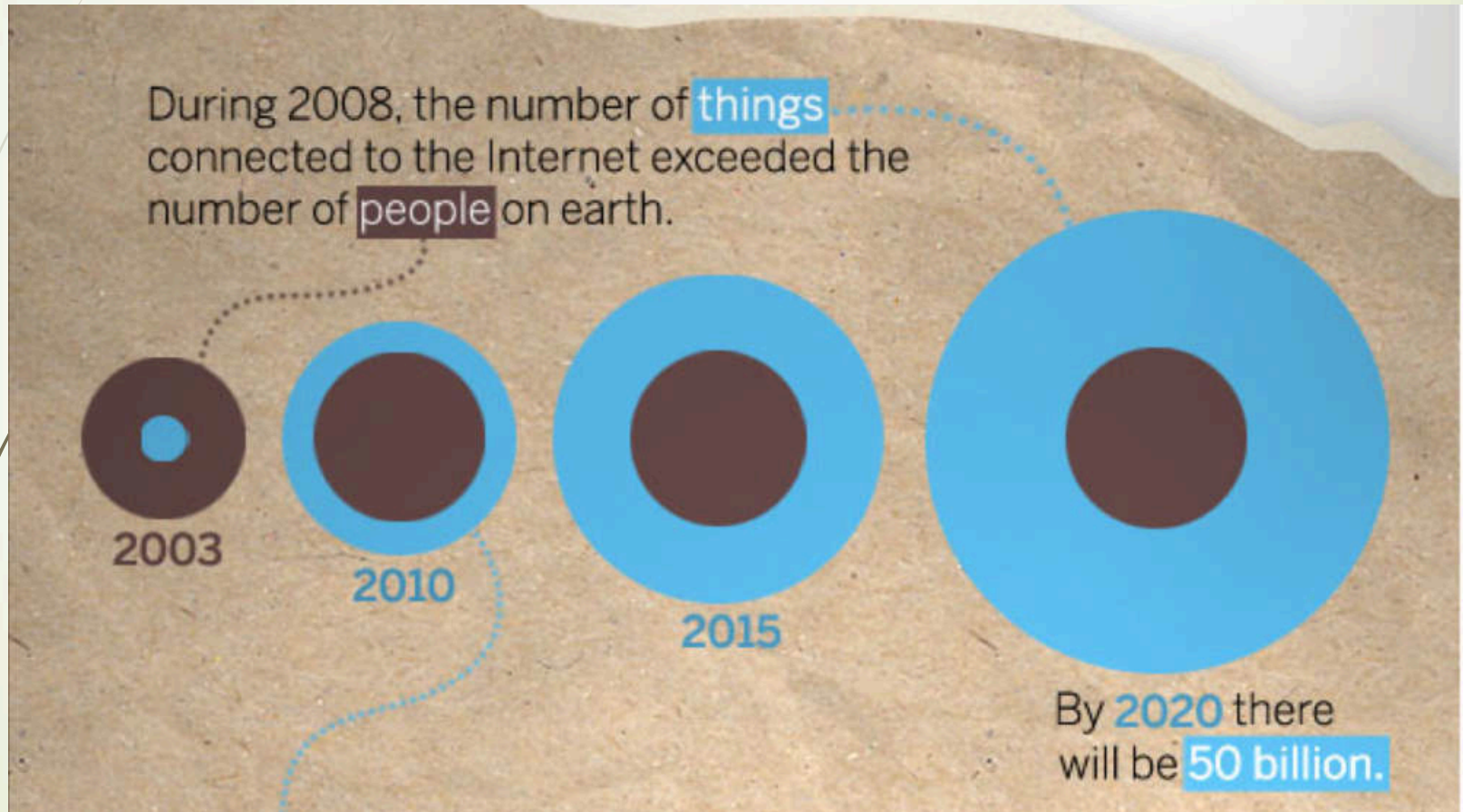
Health-care

...



# In Fact, Things are Connected to the Internet

**Q: How does this affect the network infrastructure?**

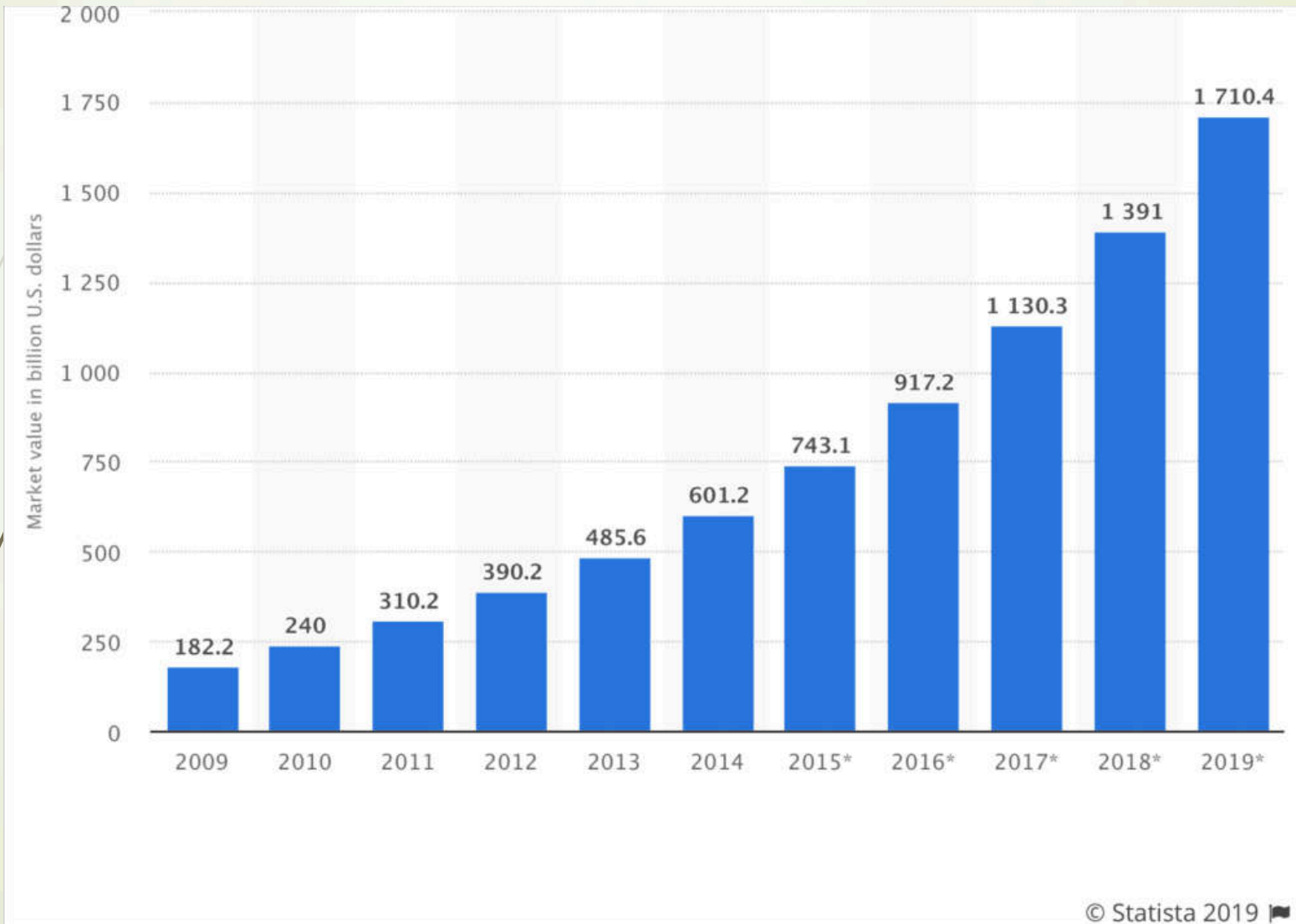


Sources: Cisco IBSG, Jim Cicconi, AT&T, Steve Leibson, Computer History Museum, CNN, University of Michigan, Fraunhofer

# Definitions of Internet-of-Things

- Extending the current Internet and providing connection, communication, and inter-networking among physical things.
- The technologies and solutions that enable integration of **real world data and services** into the **networked information technologies** are described under the umbrella term of the IoT

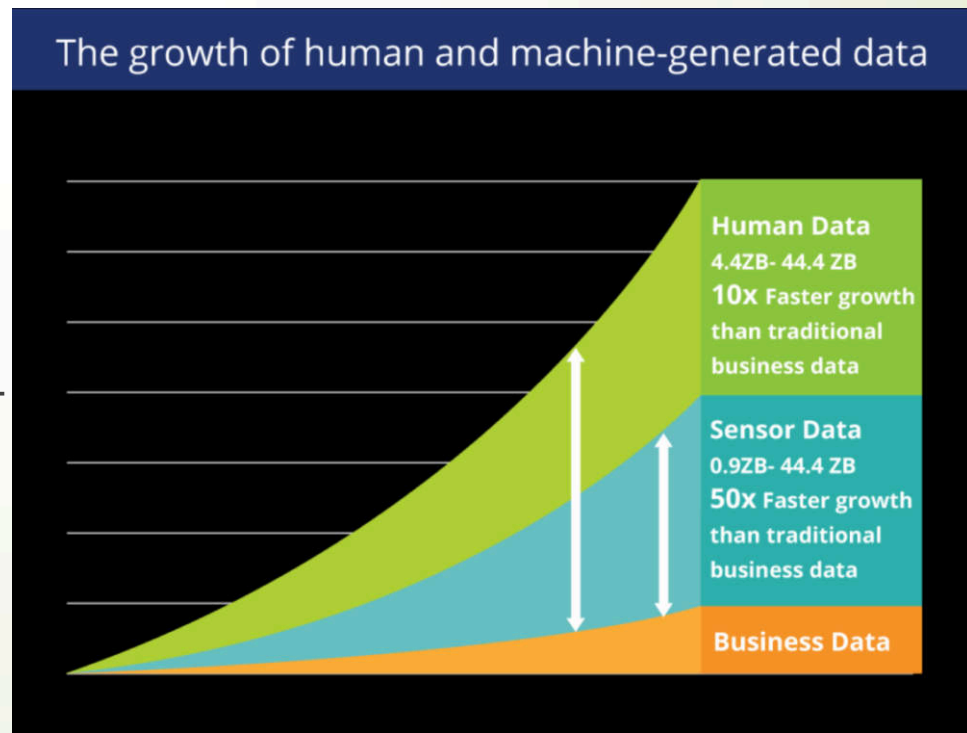
# IoT Market is Expected to Grow Fast



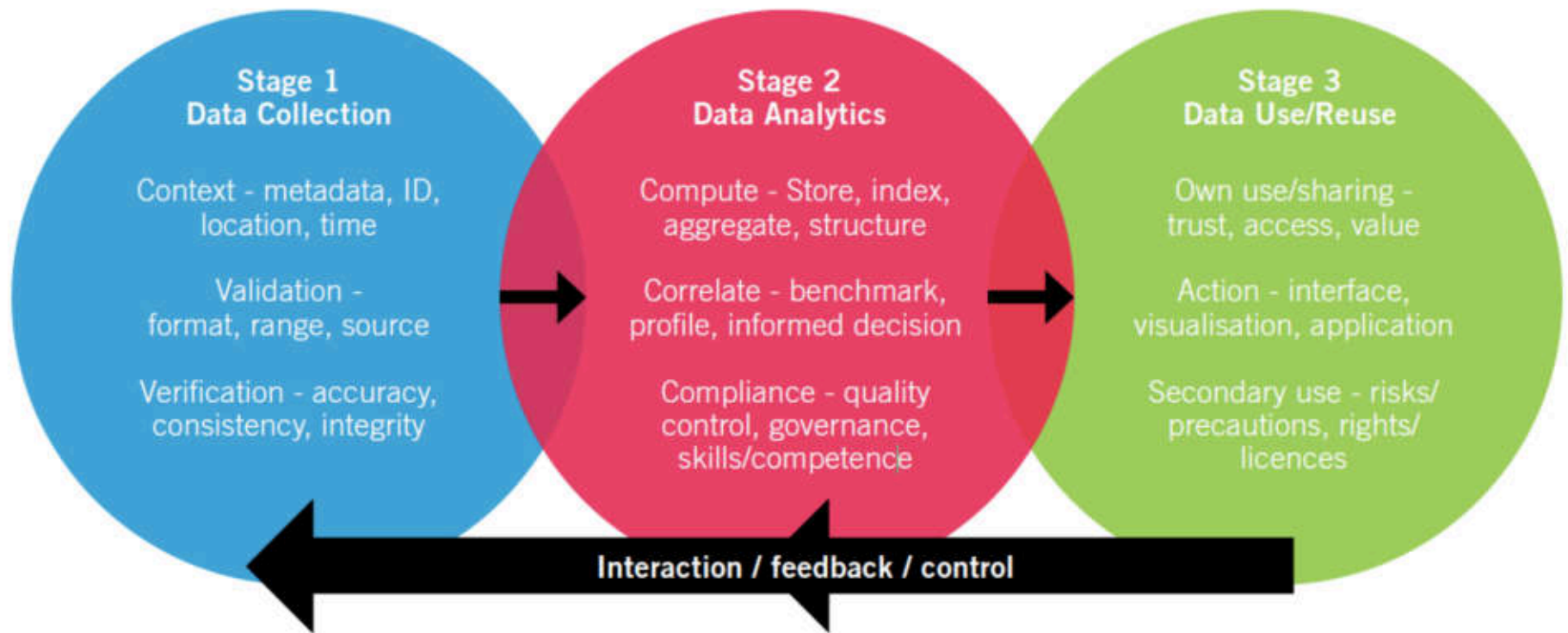
# IoT Data Generation Amount Also Increases

➤ Everyday around 20 quintillion ( $10^{18}$ ) bytes of data are produced (source: IBM), which include

- textual content (unstructured, semi-structured, structured)
- multimedia content (images, video and audio)



# Life-cycle of (IoT) Data



# Observations

- Ultimate goal is to transform the **raw data** to **insights** and actionable **knowledge** for machines and human
- Require data from multiple sources, (near-) real time analytics and visualization and/or semantic representations
- ➔ **IoT analytics!**